

A **statistical experiment** is any procedure that produces data or observations.

The **sample space**, denoted by S, is the set of all possible outcomes of a statistical experiment. The sample space depends on the problem of interest!

A **sample point** is an outcome (element) in the sample space. An **event** is a subset of the sample space.

A is contained in B, denoted by $A \subset B$, or equivalently $B \supset A = A$ is a proper subset of B. A is contained within B but not equal to B.

EVENT OPERATIONS

$A \cap A' = \emptyset, A \cap \emptyset = \emptyset, A \cup A' = S, (A')' = A, A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), A \cup B = A \cup (B \cap A'), A = (A \cap B) \cup (A \cap B')$

DE MORGAN'S LAW

$(A_1 \cup A_2 \cup \dots \cup A_n)' = A_1' \cap A_2' \cap \dots \cap A_n'.$

$(A_1 \cap A_2 \cap \dots \cap A_n)' = A_1' \cup A_2' \cup \dots \cup A_n'.$

Counting Methods

Multiplication is used when there are n different events (independent) to be performed **sequentially**.

Addition is used when the same event can be performed by k different procedure (mutually exclusive).

Permutation

The number of ways to arrange r objects out of n.

$${}^n P_r = \frac{n!}{(n-r)!} = {}^n C_r \times r P_r$$

• Permutations around a circle = $(n-1)!$

• Permutations when not all objects are distinct = $\frac{n!}{n_1!n_2!\dots n_k!}$

Combination

A selection of r objects out of n, without regard to the order.

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r} = \binom{n-1}{r} + \binom{n-1}{r-1}$$

Probability

$$\Pr(A) = \lim_{n \rightarrow \infty} f_n A$$

• An interpretation of probability is relative frequency. (Total number of times occurred / Total number of times repeated)

• If A and B are mutually exclusive events, $f_{A \cup B} = f_A + f_B$, (Just add their probability) that is, $A \cap B = \emptyset$, $P(A \cup B) = P(A) + P(B)$.

• The probability of the empty set $\emptyset$ is $P(\emptyset) = 0$.

• If $A_1, A_2, \dots, A_n$ are mutually exclusive events, that is $A_i \cap A_j = \emptyset$ for any $i \neq j$, then $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$.

• For any two events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

• If A is contained by B, then $P(A) \leq P(B)$

Conditional Probability

For any two events A and B with $P(A) > 0$, the conditional probability of B given that A has occurred is defined by

$$!!! P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)P(A|B)}{P(B)P(A|B) + P(B')P(A|B')} !!!$$

Two events are said to be **independent if and only if** $P(A \cap B) = P(A) \times P(B)$. If A and B are independent, then

• their conditional probability, $P(A|B) = P(A)$.

• $A \perp B'$, $A' \perp B$, and $A' \perp B'$.

• A and B cannot be mutually exclusive, if $P(A), P(B) > 0$

• All events are independent with S and $\emptyset$.

Law of Total Probability

$$P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(A_i)P(B|A_i)$$

• Assuming that events $A_1, \dots, A_n$ are mutually exclusive and exhaustive events. (They are **partition** of S)

Monty Hall Problem

Suppose you chose to switch, $W = \{\text{Win the car}\}$, $A = \{\text{Car is behind the door of initial pick}\}$

$$P(W) = P(A) \times P(W|A) + P(A') \times P(W|A') = \frac{1}{3}(0) + \frac{2}{3}(1) = \frac{2}{3}$$

Birthday Problem

$P(> 2 \text{ people have same birthday}) = \Pr(n) = 1 - \Pr(q)$, where $\Pr(q)$ is the probability that every person in the room has a different birthday i.e.

$$\frac{365(364) \dots (365-n+1)}{365^n}$$

Inverse Birthday Problem

$P(\text{someone has the same birthday as you}) = \Pr(n) = 1 - \Pr(q) \geq 0.5$, where $\Pr(q)$ is the probability of someone having a different birthday

$$\text{i.e. } \left(\frac{364}{365}\right)^n$$

Bayes' Theorem

Let $A_1, A_2, \dots, A_n$ be a partition of S, then for any event B and $k = 1, 2, \dots, n$,

$$P(A_k|B) = \frac{P(A_k \cap B)}{P(B)} = \frac{P(A_k)P(B|A_k)}{\sum_{i=1}^n P(A_i)P(B|A_i)} = \frac{P(A_k)P(B|A_k)}{\sum_{i=1}^n P(A_i)P(B|A_i)}$$

• Bayes Theorem can be derived using conditional probability, multiplication rule and the law of total probability.

Random Variables

• A function X, which assigns a real number to every $s \in S$ is called a random variable. **Range space**, $R_x = \{x|x = X(s), s \in S\}$

• Likewise, the set $X \in A$, for A being a subset of R , is also a subset of S: $\{s \in S : X(s) \in A\}$.

Discrete Probability Distribution

Each value of X has a certain probability, $f(x)$, and this function f is called the **probability mass function**. It must satisfy the following:

1. $0 \leq f(x_i) \leq 1$ for all $x_i \in R_x$, otherwise 0.

2. Sum of the probabilities, $\sum_{i=1}^{\infty} f(x_i) = 1$

Continuous Probability Distribution

R_x is an **interval** or a collection of intervals. The **probability density function** quantifies the probability that X is in a certain range. It must satisfy the following:

1. The probability of the continuous random variable X = to any observed value is always 0.

2. $f(x) \geq 0$ for all $x \in R_x$, otherwise 0.

3. $P(A) = 0$, but A is not necessarily $\emptyset$ or not observed. (Point 1 & 6)

4. Not necessary that $f(x) \leq 1$, can be greater than 1, as it represents density, not direct probability.

5. Sum of probabilities = $\int_{R_x} f(x) dx = 1$

6. For any a and b such that $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(x) dx$

Note: $P(A < X < B) = P(A \leq X \leq B)$

Cumulative Probability Distribution

For any random variable X, define its CDF by $F(x) = P(X \leq x)$.

For the **discrete** case, the CDF is a step function. The jumping points of the CDF forms the values of X. For any two numbers $a < b$, we have $P(a \leq X \leq b) = P(X \leq b) - P(X < a) = F(b) - F(a-)$.

For the **continuous** case, $F(x) = \int_{-\infty}^x f(t) dt$, $f(x) = \frac{d}{dx} F(x)$.

$P(a \leq X \leq b) = P(a < X < b) = F(b) - F(a)$.

Some remarks:

• $0 \leq F(x) \leq 1$

• $F(X)$ is non-decreasing i.e. $x_1 < x_2, F(x_1) \leq F(x_2)$.

• PF and CDF have a one-to-one correspondence. (Unique CDF)

• CDF must be right continuous. (Close dot then open dot, etc)

• Discrete CDF usually has limited fixed values or no function.

• Continuous CDF usually have a function or a range of values.

Expectation and Variance

For **discrete** random variable,

$$\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i) \quad V(X) = \sum_{x \in R_X} (x - \mu_X)^2 f(x).$$

For **continuous** random variable,

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{x \in R_X} x f(x) dx.$$

$$V(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx$$

Properties of Expectation and Variance

• $E(aX + b) = aE(X) + b$

• $E(X + Y) = E(X) + E(Y)$, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

• $\text{Var}(aX + b) = a^2 \times \text{Var}(X)$, $\text{Var}(X) = E(X^2) - [E(X)]^2$.

• Discrete: $E[g(X)] = \sum_{x \in R_X} g(x) f(x)$

• Continuous: $E[g(X)] = \int_{x \in R_X} g(x) f(x) dx$

• Standard Deviation ≥ 0 , $[E(S)]^2 \leq E(S^2) = \sigma^2$, S = Sample

• There exist probability distributions for which $E(X)$ do not exist, i.e.

$E(X) = \infty$

• Variance = 0, if $P(X = E(X)) = 1$

Discrete Distributions

If X assumes the values $x_1, \dots, x_n$ with equal probability, then X follows a **discrete uniform distribution**.

$$P(X = x) = \frac{1}{n} \quad \text{for } x \in \{x_1, x_2, \dots, x_n\}$$

$$E(X) = \frac{1}{k} \sum_{i=1}^k x_i \quad V(X) = \left(\frac{1}{k} \sum_{i=1}^k x_i^2 \right) - \mu_X^2$$

Bernoulli Distribution

A **Bernoulli Trial** has two possible outcomes. The number of successes in a Bernoulli trial, $X \sim \text{Bernoulli}(p)$.

A **Bernoulli process** consists of a sequence of repeatedly performed independent and identically distributed (i.i.d) Bernoulli trials.

$$f_X(x) = p^x(1-p)^{1-x}, \quad \text{for } x = 0 \text{ or } 1$$

$$E(X) = p \quad V(X) = p(1-p)$$

Remark: The collection of the distributions that are determined by one or more unknown **parameters** is called a **family of probability distributions**.

Binomial Distribution

Let X be number of successes in n **Bernoulli trials** where

1. the probability of success for each trial is the same p,

2. the trials are independent then X follows a binomial distribution, $X \sim B(n, p)$.

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \text{for } x = 0, 1, \dots, n$$

$$E(X) = np \quad V(X) = np(1-p)$$

Remark: Bernoulli is a special case of Binomial, where $n = 1$. Bernoulli trial just tells you the probability of success and failure of 1 trial. If X, Y are i.i.d RV, then $X + Y \sim B(2n, p)$. As p increases and n remains the same, the probability mass function starts to skew left. (Peak shift right)

Negative Binomial Distribution

Let X be number of i.i.d Bernoulli trials until the kth success occurs, p is the probability of 1 success, $X \sim \text{NB}(k, p)$.

$$P(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad \text{for } x = 0, 1, \dots, n$$

$$E(X) = \frac{k}{p} \quad V(X) = \frac{(1-p)k}{p^2}$$

Remark: Geometric Distribution is a special case of Negative Binomial Distribution, where $k = 1$. Trials needed till first success occur.

Poisson Distribution

Let X be number of occurrences in a fixed period of time or fixed region, $X \sim \text{Poisson}(\lambda)$ where $\lambda > 0$ is the expected number of occurrences.

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad E(X) = \lambda = V(X)$$

When lambda increases, we can see that the mode shifts right, shape of the PMF becomes more symmetrical.

Poisson Process is a continuous time process. The properties of a Poisson Process with rate parameter α :

1. An interval of length T has $E(X) = \alpha T$

2. There are no simultaneous occurrences

3. The number of occurrences in disjoint time intervals are independent

Remark: If X, Y are i.i.d RVs, then $X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$

Qn usually give λ but you have to scaled accordingly to ensure the time period is the same before using the formula above. 1 day: λ , 2 day: 2λ

Poisson approximation of Binomial Distribution

Suppose $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np$ remains a constant. Then, $X \sim \text{Poisson}(np)$. This approximation is valid when $n \geq 20$ and $p \leq 0.05$, or if $n \geq 100$ and $np \leq 10$. Use the same formula but replace λ with np .

Continuous Distribution

A random variable X is said to follow a uniform distribution over the interval (a,b) if its PDF is given by:

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b \quad E(X) = \frac{a+b}{2} \quad V(X) = \frac{(b-a)^2}{12}$$

and its CDF is given by:

$$F_X(x) = 0 \text{ if } x < a \quad F_X(x) = \frac{x-a}{b-a} \text{ if } a \leq x \leq b \quad F_X(x) = 1 \text{ if } x > b$$

Exponential Distribution

The exponential distribution is the continuous counterpart to the geometric distribution. It is often used to model the waiting time to the first success in continuous time.

$$f_X(x) = \lambda e^{-\lambda x}, \quad \text{if } x \geq 0 \quad f_X(x) = 0, \quad \text{if } x < 0$$

$$E(X) = \frac{1}{\lambda} \text{ and } V(X) = \frac{1}{\lambda^2}$$

• y-intercept is lambda!! Strictly decreasing function. As λ increase, graph becomes more asymptotic at both x and y-axis.

The CDF of $X \sim \text{Exp}(\lambda)$ is given by:

$$F_X(x) = 1 - e^{-\lambda x} \text{ if } x \geq 0 \quad F_X(x) = 0 \text{ if } x < 0$$

• For $x > 0$, $P(X > x) = e^{-\lambda x}$. Alternative form, λ can be sub as $1/\mu$

• Sometimes they like to test you on finding the probability of an event first, then perform Binomial Distribution.

• **Memory-less:** For $\lambda > 0$ and any two positive numbers s and t, we have $P(X > s+t | X > s) = P(X > t)$. (Only Geometric and Exponential)

Normal Distribution

The PDF of normal distribution is positive over the whole real line, **symmetric** about $x = \mu$, and **bell-shaped**. Its PDF is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty.$$

Its CDF is given by

$$F_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Standardization:

$$X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Properties:

• PDF of Z: Replace $(x - \mu)^2 / \sigma^2$ with Z, σ with 1

• $\Phi(z) = P(Z \leq z) = P(Z \geq -z) = P(-Z \leq z) = 1 - F(-z) = 1 - \Phi(-z)$

• $P(Z \geq 0) = P(Z \leq 0) = 0.5$.

• If $Z \sim N(0, 1)$ then $\sigma Z + \mu \sim N(\mu, \sigma^2)$

• $P(|Z| < z) = 2F(z) - 1$

- By symmetry, $-Z \sim N(0, 1)$, $E(X^{2k+1}) = 0$ for $k = 0, 1, \dots, n$
 - For e.g. 95th percentile = $P(Z \geq z_{0.05}) = 0.05$
- A **z-score** tells you how many standard deviations a data point is from the mean.

Normal Approximation to Binomial

Let $X \sim B(n, p)$. $E(X) = np$ and $V(X) = np(1 - p)$. Then as $n \rightarrow \infty$,

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} \sim N(0,1)$$

This approximation is good when $np > 5$ and $n(1 - p) > 5$.

Continuity Correction

- Continuity correction is a tweak made when approximating a discrete distribution (like a binomial distribution) with a continuous distribution (like a normal distribution).
- $P(X = k) \approx P(k - 1/2 < X < k + 1/2)$
- $P(a \leq X \leq b) \approx P(a - 1/2 < X < b + 1/2)$
- $P(a < X \leq b) \approx P(a + 1/2 < X \leq b + 1/2)$
- $P(a \leq X < b) \approx P(a - 1/2 < X < b - 1/2)$
- $P(a < X < b) \approx P(a + 1/2 < X < b - 1/2)$
- $P(X \leq c) = P(0 \leq X \leq c) \approx P(-1/2 < X < c + 1/2)$
- $P(X > c) = P(c < X \leq n) \approx P(c + 1/2 < X < n + 1/2)$

Sampling Distribution

The number of possible **observations** of an experiment is called a population. A **sample** is a subset of a population $\rightarrow$ infinite if the # of observation is infinite.

Random Sampling

A random sample is **unbiased** and **models** the population.

(Finite) Simple random sample: Every subset of n observations has the same probability of being selected, $\binom{N}{n}$ possible samples. **Note:** Sampling with replacement is infinite and random only if (1) P(Selected) is same (2) Draws are independent

(Infinite) Let X be a random variable with certain probability distribution $f_X(x)$. Let $X_1, X_2, \dots, X_n$ be n independent random variables each having the same distribution as X. Then $(X_1, X_2, \dots, X_n)$ is called a random sample of size n. The **joint probability function** is given by:

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

where $f_X(x)$ is the probability function of the population.

Sample Distribution of Sample Mean

A function of $(X_1, \dots, X_n)$ is called a **statistic**. There cannot be any unknown variables. The sampling distribution of the **sample mean** has

$$E(\bar{X}) = \mu_x, \quad V(\bar{X}) = \frac{\sigma_x^2}{n}$$

For an **infinite** population, when $n \rightarrow \infty$, $V(\bar{X}) \rightarrow 0$. The accuracy of $\bar{X}$ as an estimator of μ_X keeps improving. This is known as the **law of big numbers**, $P(|\bar{X} - \mu| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$. The **standard error**/standard deviation of $\bar{X}$ describes how much $\bar{x}$ varies from sample to sample of size n. Standard Error $\neq$ Margin of Error.

Central Limit Theorem

If $\bar{X}$ is the **mean of a random sample** (i.i.d) of size n taken from a population having mean μ and finite variance σ^2 , then, as $n \rightarrow \infty$,

$$\bar{X} \rightarrow N\left(\mu, \frac{\sigma^2}{n}\right)$$

Chi-Squared Distribution

Let $Z_1, \dots, Z_n$ be i.i.d. standard normal random variables. Then,

$$\sum_{i=1}^n Z_i^2 \sim \chi^2(n)$$

with **n degrees** of freedom.

Properties

1. If $Y \sim \chi^2(n)$, then $E(Y) = n$ and $V(Y) = 2n$.
2. For large n, $\chi^2(n)$ is approximately $N(n, 2n)$.
3. Linearity of freedom. $Y_1 + Y_2 \sim \chi^2(m + n)$. (Both are independent)
4. Right skewed. (Long right tail)
5. $P(Y > \chi^2(n; \alpha)) = \alpha$, (Right tail area is α & chi-score is χ_{α}^2)

Sampling Distribution of Sample Variance

Let S^2 be the variance of a random sample taken from a normal population with variance σ^2 , then (Can derive S^2 formula here as well)

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2}$$

has a χ^2 distribution with $n - 1$ degrees of freedom. It can be shown that $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n(\bar{X})^2$, thus $E(S^2) = \sigma^2$. But $E(S) \neq \sigma$.

T Distribution

Suppose $Z \sim N(0,1)$ and $U \sim \chi^2(n)$. If Z and U are independent, then $T = \frac{Z}{\sqrt{U/n}} \sim t_n$ follows the t-distribution with **n degrees** of freedom.

$$E(T) = 0 \quad \& \quad V(T) = \frac{n}{n-2} \quad \text{for } n > 2$$

The t-distribution approaches $N(0,1)$ as the parameter $n \rightarrow \infty$. When $n \geq 30$, we can replace it by $N(0,1)$. Symmetric about vertical axis, like standard normal. $P(T > t_{n,\alpha}) = \alpha$ (Right tail area is α & t-score is t_{α})
If $X_1, \dots, X_n$ are independent and identically distributed normal random variables with mean μ and variance σ^2 , then $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ follows a t-distribution with $n-1$ degrees of freedom. (S = Sample Variance)

F Distribution

Suppose $U \sim \chi^2(m)$ and $V \sim \chi^2(n)$ are **independent**. Then the distribution of the random variable

$$F = \frac{U/(m)}{V/(n)} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(m, n)$$

is called a F-distribution with (m,n) degrees of freedom.

$$E(X) = \frac{n}{n-2} \text{ for } n > 2, \quad V(X) = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)} \text{ for } n > 4$$

Properties

- If $F \sim F(m, n)$, then $1/F \sim F(n, m)$
- $P(F > F(m, n; \alpha)) = \alpha$ where $F \sim F(m, n)$
- $F(m, n; 1 - \alpha) = 1/F(n, m; \alpha)$ (**NOTE** the swap in degrees of freedom)

Estimation of Population Parameters

A **point estimate** estimates the population parameter. A **point estimator** is the method/formula that describes this calculation. An **interval estimation** forms an interval in which the parameter is estimated to lie.

Unbiased Estimator

Let $\hat{\theta}$ be an estimator of θ . If $E(\hat{\theta}) = \theta$, we say that $\hat{\theta}$ is an unbiased estimator of θ . As $V(\hat{\theta})$ decreases, the estimation gets better. An unbiased estimator has mean value equals to the true value of the parameter.

Maximum Error of Estimate

$E = \bar{X} - \mu$ measures the difference between the estimator and the parameter value. The maximum error, $E \sim N(0, 1)$, where

$$P\left(|\bar{X} - \mu| \leq z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Confidence Interval

An interval (a,b) in which you are fairly certain the parameter of interest lies in. This can be quantified by the degree of confidence, $P(a < \mu < b) = 1 - \alpha$. The **general formula** is given by $\bar{X} \pm E_0$.

Hypothesis Testing

Test statistic quantify how unlikely it is to observe the sample, given that the null hypothesis is true.

P-value is the probability of obtaining a test statistic **at least as extreme** ($\leq$ or $\geq$) than the observed sample value, given H_0 is true.

Types of Errors

- Type 1 error - We reject H_0 when H_0 is true.
- Type 2 error - We do not reject H_0 when H_0 is false.

Type 1 error is also known as the **level of significance**, $\alpha = P(\text{Type 1 error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$.

Let $\beta = P(\text{Type 2 error}) = P(\text{Do not reject } H_0 \mid H_0 \text{ is false})$. Then, $1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ is false})$ is the **power of the test**, a.k.a. **true negative rate**. If the α decreases, likelihood of not rejecting H_0 increases $\Rightarrow$ power of the test decreases.

Known Population Variance (Single Sample)

We consider the case where (1) n is sufficiently **large**, or (2) the underlying distribution is **normal**.

1. Perform standardization, $z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$. (z used in part 4)

2. Confidence Interval: $\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

3. $H_1: \mu \neq \mu_0$, the rejection region is $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$

4. For two sided tests, reject H_0 if $2P(Z > |z|) < \alpha$.

Unknown Population Variance (Single Sample)

Underlying distribution is **normal**, and sample size is **small**.

1. Perform standardization, $t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$

2. For two sided tests, reject H_0 if $2P(T > |t|) < \alpha$

3. Confidence Interval: $\bar{x} \pm t_{n-1, \alpha/2} \cdot \frac{s}{\sqrt{n}}$

Note: If $n \geq 30$, then $T \sim N(0, 1)$, Replace $t_{n-1, \alpha/2}$ with $z_{\alpha/2}$.

Two Sided Tests and Confidence Intervals

Two sided hypothesis test is equivalent to finding a $100(1 - \alpha)\%$ confidence interval for μ . If the confidence interval contains μ_0 , H_0 will not be rejected at level of significance, α .

Two Independent Samples (Known & Unequal Variances)

1. The population variances are **known** and **not** the same: $\sigma_1^2 \neq \sigma_2^2$
2. Two populations are **normal** or **large**

$$E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2 = \delta, \quad V(\bar{X} - \bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

$$\text{Confidence Interval for } \mu_1 - \mu_2 = (\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

For the null hypothesis $H_0: \mu_1 - \mu_2 = \delta_0$, the test statistics is given by

$$Z = \frac{(\bar{X} - \bar{Y}) - \delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

Two Independent Samples (Large with Unknown Variances)

1. The population variances are **unknown** and **not** the same: $\sigma_1^2 \neq \sigma_2^2$
2. Both samples are large: $n_1 \geq 30, n_2 \geq 30$.

C.I & Test Statistic same as (Known & Unequal Variances), just replace σ with s .

Two Independent Samples (Small with Equal Variances)

1. The population variances are **unknown** and the **same**: $\sigma_1^2 = \sigma_2^2 = \sigma^2$.
2. Both samples are **small**: $n_1 < 30, n_2 < 30$.
3. Both populations are **normally distributed**.

Pooled Estimator (Equal variance if S_1/S_2 is between $1/2$ and 2)

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

When we estimate σ^2 using S_p^2 , the resulting statistic transform from

$$T = \frac{(\bar{X} - \bar{Y}) - \delta_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1 + n_2 - 2}$$

$N(0,1)$ to a t-distribution with degrees of freedom $n_1 + n_2 - 2$. Therefore,

C.I for Small with Equal Variance: $(\bar{X} - \bar{Y}) \pm t_{n_1 + n_2 - 2, \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

NOTE: For **large** samples, we can replace $t_{n_1 + n_2 - 2, \alpha/2} S_p$ by $z_{\alpha/2}$ in the previous formula. No change to Test Statistics formula $\rightarrow \sim N(0,1)$.

Paired Data (Two Dependent Sample X & Y)

1. (X_i, Y_i) and (X_j, Y_j) are independent for any $i \neq j$.
2. For matched pairs, define $D_i = X_i - Y_i$, $\mu_D = \mu_1 - \mu_2$.
3. All techniques derived for a single population can now be employed.

If $n < 30$ & population is **normally** distributed, $\bar{D} \pm t_{n-1, \alpha/2} \cdot \frac{S_D}{\sqrt{n}}$

For the null hypothesis $H_0: \mu_D = \mu_{D_0}$, the test statistics is given by

$$T = \frac{\bar{D} - \mu_{D_0}}{S_D/\sqrt{n}} \sim t_{n-1}$$

For large samples, we can replace $t_{n-1, \alpha/2}$ by $z_{\alpha/2}$.

Tips & Tricks

1. When asked to find CDF of a continuous variable, sub x as t and integrate w.r.t t from the lower bound of the range to x. If Qn gives multiple ranges e.g. $0 \leq x < 1$ & $1 \leq x < 2$. Do the usual integration but when integrating $1 \leq x < 2$, **remember to add** F(1) to the final integral.
2. If X is translated e.g $Z = 3X + 2$, the $f(z)$ probability stays the same as $f(x)$.
3. profit = $600t - 50X^2 \rightarrow E(\text{profit}) = 600t - 50E(X^2)$, then use $V(X) = E(X^2) - [E(X)]^2$. This question was on Poisson where $\lambda = 0.8t$, so just sub $E(X)$ as $0.8t$ to find $E(X^2)$ and differentiate it w.r.t t. Stationary Points: $\frac{dy}{dx} = 0$.
4. Usually, regardless of what distribution, when calculating expectation, you should add in $g(x)$ to your summation or integration.
5. $P\left(X < -\frac{1}{4} \text{ or } X > \frac{1}{4}\right) = 1 - P\left(-\frac{1}{4} \leq X \leq \frac{1}{4}\right)$
6. $E(X) = \sum_{k=1}^{\infty} P(X \geq k)$, CLT only applies to sample mean.
7. $P(X \leq x) = \sum_{n=1}^x p q^{n-1} = p \cdot \frac{1-q^x}{1-q} = 1 - q^x$ (CDF of Geometric Distribution using sum of **first n terms** of Geometric Series)
10. **IMPORTANT:** If there is a **common probability ratio** between Bernoulli trials, E.g. find P(A Win) but there are infinite rounds, it is the Sum of **infinite** Geometric Series. Sum = $\frac{a}{1-r}$
9. $P(X > x) = P(\min\{X_1, X_2, \dots, X_n\} > x) = P(X_1 > x, X_2 > x, \dots, X_n > x) = P(X_1 > x)P(X_2 > x) \cdots P(X_n > x) = e^{-\lambda x} e^{-\lambda x} \cdots e^{-\lambda x} = e^{-n\lambda x}$. This implies that $X \sim \text{Exp}(n\lambda)$.
10. Whenever an approximation is made to Binomial, Expectation and Variance will follow Binomial Distribution.
11. $P(A \text{ before } B) = \frac{P(A)}{P(A) + P(B)}$
12. Both the sample mean $\bar{X}$ and X_i are unbiased estimators for μ .
13. **Standard Error** = $\frac{\sigma}{\sqrt{n}}$, Margin of Error (C.I) = $z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$
14. It is possible that $s^2 > \sigma^2$ & $s^2 < \sigma^2$ as they are observed sample variances.
15. Sometimes if probability is not given, try use **arbitrary value** for the probability of heads, then work out the algebra.
16. When asked to **find PDF of continuous RV** after transformation, e.g. Let $Y = \frac{1}{1-X}$. Then, $F(Y) = P(Y \leq y) = P(\frac{1}{1-X} \leq y) = P(X \leq 1 - \frac{1}{y})$.

This is CDF so integrate the original PDF, X, lower bound to $1 - 1/y$. Differentiate the final CDF of Y to get PDF of Y. Replace X with the new X based on the formula for Y (Refactor first).

17. For joint probability function, Discrete = Double summation, Continuous = Double integral. The inner one is the one that relies on the outer (E.g. y is formulated in terms of x).